

The deck is good. It describes a system we **didn't build**.

Every technical claim on the slides was checked against our actual repository. The structure, the visual identity and the real app screenshots are strong — keep them. But eight claims name technology that is nowhere in our code, and finale judges have the prototype in front of them.

8**False tech claims**

Named on slides, absent from the code

7**Errors & typos**

Including INAPH and NADRES, on the references slide

4**Unsourced numbers**

Need a citation or a softer word

4**Additions that win marks**

Exact copy provided, ready to paste

THE ONE THAT MATTERS MOST

We claim **DBSCAN** for outbreak detection. We actually built a **Kulldorff space-time scan statistic with Poisson p-values** — peer-reviewed epidemiology that outputs $p = 0.003$ instead of "these dots look close together". We replaced our strongest differentiator with a weaker buzzword. Fixing this costs one sentence and makes the deck both more impressive and true.

Claimed on the slides vs what is actually in the repository

A finale judge who asks "show me the XGBoost model" or "where is your Kubernetes cluster?" gets no answer today. One unanswerable question puts every other claim in doubt.

CLAIMED ON THE DECK	WHAT WE ACTUALLY BUILT	EVIDENCE
XGBoost ranks urgency (3 places)	Transparent weighted sign-scoring over a hot-reloadable KB	no xgboost in repo
DBSCAN finds clusters	Kulldorff space-time scan, Poisson p-value	no sklearn in repo
PyTorch · ONNX	TensorFlow / Keras EfficientNetV2	not installed
Flutter · React.js PWA	Vanilla-JS PWA, no build step	no react, no flutter
Node.js backend	FastAPI (Python) only	not present
Redis · AWS S3 · Kubernetes (EKS) · Prometheus	SQLite / Postgres on Render + ngrok	none deployed
8 species incl. Dog & Horse	6 species; no dog or horse disease at all	diseases.yaml
5+ years of disease trends	24 months seeded	seed.py months=24

Paste this instead (slide 2, Innovation)

- **Outbreak Radar:** Kulldorff space-time scan statistic — observed vs expected cases with a Poisson p-value, so every cluster alert is statistically defensible, not a dot-density guess.
- **Explainable Triage:** transparent weighted sign-scoring over a hot-reloadable knowledge base — the department edits the rules; no black box.
- **Pashu Lens:** EfficientNetV2 breed identification with a subject gate that refuses non-cattle instead of guessing.
- **Voice-First Reporting:** a guided interview in Hindi/Marathi — milk, feed, fever, lumps — producing the same structured case a form does.

Paste this instead (slide 3, Stack)

- **Frontend:** framework-free PWA — installs on a ₹5,000 Android, works offline, no app store
 - **Backend:** FastAPI + SQLAlchemy, 53 typed endpoints, auto OpenAPI
 - **Data:** PostgreSQL / PostGIS (SQLite for zero-setup demo)
 - **AI/ML:** TensorFlow EfficientNetV2 · Kulldorff scan · SEIR forecast · Gemini for free-form voice
 - **Deploy:** any MeitY-empanelled cloud; runs today on a single instance
- Naming a small, real stack reads as engineering maturity. A long aspirational stack reads as a wish list.

Small mistakes, on the slides where accuracy is the whole point

SLIDE	WRONG	CORRECT	DONE
3, 4	INAF	INAPH — Information Network for Animal Productivity & Health	☐
6	NADE-RS	NADRES — National Animal Disease Referral Expert System	☐
5, 6	HCKATHON 2026	HACKATHON 2026	☐
2	AI Traige (step 02)	AI Triage	☐
3	"Bharat Pashudhan (INAF)" — text cut off, bracket missing	Widen the box; it is overflowing	☐
2 vs 4	"18+ farmers" vs "20-farmer pilot"	Pick one number and use it everywhere	☐
2	"3.3K+ images" and "40k+ images" on the same slide	One figure, or drop it	☐

Why these two hurt most. INAPH and NADRES are misspelled on the slide titled *Research and References*, in front of judges from the animal-husbandry department who use those systems daily. It is the cheapest possible credibility loss.

⚠ Claims that need a source, or a softer word

CLAIM	THE PROBLEM, AND WHAT TO DO
"95%+ accurate model"	We have no measured accuracy for the 51-breed model. Either produce a confusion matrix before the finale, or write "trained on 50 Indian breeds with subject + quality gates".
"43K+ records of symptoms"	Name the dataset, or remove it. A judge asking "which dataset?" must get an answer.
"₹20,000 Crore annual loss (FMD + Mastitis)"	A good, quotable number — but none of your 7 references supports it . Add the citation as reference 8.
"20-farmer pilot completed / underway"	"completed / underway" reads as hedging, and today it is neither. See the box below.

✅ Turn the weakest claim into the strongest moment

The live enrolment wall is now built: a farmer registers on a phone and appears on the district dashboard within seconds, counting towards a target of 20.

Register 20 real farmers before or during the finale. Then "20-farmer pilot completed" is simply true, and you can put the counter on screen while you say it. A claim the judges watch arriving beats any bullet point.

What no other team will be able to show

1 • Put the PS's own outcomes on the slide, with a number

The problem statement names six expected outcomes. Most teams promise them; nobody measures them. We now can — the platform records when the animal fell ill, not just when it was reported.

Reporting delay: 6.4 days → 1.5 days. *Median onset-to-report, measured inside the platform across 237 cases as adoption grows.*

Goes on slide 5 (Impact). It answers "how do you know it works?" — the question most finalists cannot answer. Say plainly that the pilot data is simulated; the *measurement* is real and runs live.

2 • Stop looking like a rival. Be the layer on top.

The comparison table marks us green on all ten rows. All-green tables read as self-serving and invite attack.

1962 treats one animal at a time. Bharat Pashudhan holds the lifetime record. Neither turns those into village-level early warning — that is our layer. *We read their data by Pashu Aadhaar; we never ask a farmer to register an animal twice.*

Then deliberately concede two rows where **they** win — the national animal registry, and 1962's MVU fleet. A team that concedes something is trusted on everything else.

3 • Lead with the safety rule — it is missing entirely

In a MedTech/HealthTech theme, the liability question is coming. Answer it before it is asked.

The system outputs a suspected category, a severity band and a route — never a diagnosis. *Diagnosis and treatment remain with the registered veterinarian. That rule is enforced in the architecture, not in a disclaimer.*

One line on slide 2 or 5. It converts the biggest risk in the theme into a design credential.

4 • Frame the disease count — and offer a live demo

"17+ diseases" sounds thin on its own. Against the statutory list it is a roadmap.

13 of the 83 diseases notifiable under the Prevention & Control of Infectious and Contagious Diseases in Animals Act, 2009, plus 4 production diseases *farmers actually ring about — mastitis, black quarter, mange, enterotoxaemia. The knowledge base is hot-reloadable: adding a disease is 20 lines of YAML and one click.*

Then offer to add a disease live, on the judges' request. No other team can do that, and it proves the department would own the rules, not us.

Every change, on the slide it belongs to

SMART INDIA HACKATHON 2026

TITLE PAGE

- Problem Statement ID – SIH26128
- Problem Statement Title- Efficient systems for early detection, prevention, and management of livestock diseases and animal health issues.
- Theme- MedTech / BioTech / HealthTech
- PS Category - Software
- Team ID - 152605
- Team Name (Registered on portal)- MadhavX-Coders

1 · Title

- "TITLE PAGE" as a heading adds nothing – replace with the one-line pitch
- Add a QR to the live app: "scan and file a report yourself"
- Underlined wrapped title looks untidy – remove the underline

PROPOSED SOLUTION: Pashu Aarogya

PashuAarogya is an offline-capable livestock surveillance platform using rule based + AI triage (43K+ records of symptoms, 3.3K+ images, 17+ diseases) and geospatial mapping with 5+ years of disease trends + weather data for early outbreak detection and faster veterinary response.

INNOVATION:

- AI Triage & Vision:** XGBoost ranks case urgency as High, Medium or Low.
- Outbreak Radar:** DBSCAN identifies emerging regional disease clusters.
- Voice-First Reporting:** Symptoms, milk reduction & advisories.
- Anomaly Flagging:** Flags unknown symptom patterns for Vet + Lab investigation.

UNIQUENESS:

- Live Field-Tested:** Validated on-ground by 18+ registered farmers to prove rural usability.
- Milk Withdrawal Alerts:** Pauses milk supply post-treatment for food safety.
- Breed Intelligence:** 95%+ accurate model (40k+ images) enabling breed-specific risk scoring.
- 8 Species Ready:** Cattle, Buffalo, Goat, Sheep, Poultry, Pig, Dog & Horse.
- SLA Escalation:** High-risk cases reach the right Vet on time.

PASHU AAROGYA AI PLATFORM

DATA INPUT LAYER

- Symptoms & Mortality
- Vaccination Records & Treatment History
- Laboratory Results
- Weather & Historical Disease Data
- Location & Animal Details

AI & ANALYTICS ENGINE

- Disease & Severity Triage (XGBoost Multi-Class, CNN Image Processing)
- Risk Scoring (Symptoms + Image + History + Breed Location)
- Outbreak Cluster Detection (Detects abnormal case clusters)
- Explainable Risk Factors (Shows key risk factors)
- Recommended Next Action (Targeted intervention)

OUTPUT & ACTIONS

- Early Warning Alerts
- Risk Map & Hotspot
- Veterinary Case Assignment
- Sample & Lab Referral
- Milk Withdrawal Alerts
- Reports & Dashboards

RESPONSE & ACTION WORKFLOW

Report → AI Triage → Risk Score → Verify/Verify → Lab Confirm → Treat & Contain → Follow-up & Learn

Pashu Aarogya TECHNICAL APPROACH

Technology Stack:

- Frontend: Flutter, React.js, PWA
- Backend: Node.js, FastAPI, JWT Auth.
- Database: PostgreSQL + PostGIS, Redis(Cache)
- AI/ML: PyTorch, ONNX, XGBoost, DBSCAN
- Storage: AWS S3
- Security: OAuth2/JWT, HTTPS/TLS 1.3, Input Validation, Encryption
- Deployment: Kubernetes(AWS EKS)

TECHNOLOGY & TOOLS

Flutter React Node.js Python FastAPI Google Colab
Kubernetes Redis PostgreSQL AWS S3 Prometheus Scikit-learn

OFFLINE CAPABILITY

- Data Capture without Internet
- Local Storage on device
- Auto Sync when network is available

EXTERNAL INTEGRATIONS

- SMS / Email Gateway
- IVR / Voice Support
- Weather APIs
- Bharat Pashudhan (INAF)
- Other Govt. Systems

3 · Technical Approach

- Replace the whole stack list with the real one (page 2)
- "Bharat Pashudhan (INAF)" – text cut off, and INAF → INAPH
- Remove Kubernetes / Redis / S3 / Prometheus logos we don't run
- Keep the architecture diagram – it is genuinely good

Pashu Aarogya FEASIBILITY & VIABILITY

FEASIBILITY

- AI-ready: XGBoost + image models for triage & risk scoring.
- Offline-first: Capture data without internet; sync when connected
- Secure: Role-based access, encryption & verified records.

VIABILITY

- Simple multilingual app for farmers & officials.
- Voice-first and offline access improves usability.
- Farmer → Field Worker → Vet → Government workflow.

PRACTICAL IMPLEMENTATION

- 20-farmer field pilot completed / underway.
- Works offline for remote areas with seamless sync.
- Easy-to-use app/UI for effortless adoption.

POTENTIAL CHALLENGES & RISKS

- Connectivity:** Patchy rural networks may disrupt updates.
- Adoption:** Farmers may hesitate to switch or use new technology.
- Data Privacy:** Sensitive farm & animal health data need strict protection.

STRATEGIES FOR OVERCOMING CHALLENGES

- Edge-computing with local SQLite storage and differential sync via 2G/SMS fallback.
- Zero-friction AI voice UI and grassroots onboarding via local Panchayats and Paravets.
- Data protection through encryption + RBAC + secure authentication.

COST & SUSTAINABILITY MODEL

Cost-Efficient Operations: Running AI cuts expensive cloud server costs.

Zero New Hardware: Integrates directly with Bharat Pashudhan (INAF), reusing government vets and field workers simply use their existing devices.

DEPLOYMENT & SCALING STRATEGY

Scalability Pipeline: Modular architecture that can scale from Block → District → State.

Cloud Agnostic: Can be hosted on any Melt+empannelled cloud environment.

4 · Feasibility & Viability

- "AI-ready: XGBoost" → the real method
- "20-farmer pilot completed / underway" – decide, then state it
- INAF → INAPH in the cost box
- Add DPDP Act 2023 consent to the Data Privacy row

Pashu Aarogya IMPACT AND BENEFITS

POTENTIAL IMPACT ON TARGET AUDIENCE

PASHUPALAK

- Early disease alerts **reduce** livestock losses.
- Zero-friction reporting via voice-AI for semi-literate users.

FIELD WORKERS

- Offline-first case reporting.
- Logs epidemiological data in zero-network zones with auto-differential sync.

GOVERNMENT/OFFICIALS

- Village / Block / District risk monitoring.
- Faster hotspot detection & resource planning.

COMMUNITY

- Multilingual advisories support prevention.
- Faster coordination across nearby villages.

STATE-LEVEL IMPACT

- Stronger animal-health surveillance.
- Real-time spatial heatmaps to deploy targeted FMD/Bruceellosis vaccination drives

BENEFITS OF THE SOLUTION

SOCIAL

- Zoonotic Spillover Defence:** Early spatial clustering acts as a shield against animal-to-human disease transmission.

ECONOMIC

- Combats the ₹20,000 Crore annual economic loss caused by FMD and Mastitis.
- Protects agricultural GDP by securing daily milk yields and international trade compliance.

ENVIRONMENT

- Mitigates Antimicrobial Resistance (AMR):** Automated Milk Withdrawal Alerts prevent antibiotic residues from entering the human food chain.

One platform connecting Farmer → Field Worker → Vet → Government

5 · Impact and Benefits

- "HCKATHON" typo, top right
- ₹20,000 Cr needs a reference
- Add the reporting-delay number – 6.4 → 1.5 days
- Add zone-wide advisories: 18 villages warned, not just the one that reported

Pashu Aarogya RESEARCH AND REFERENCES

References:

- GOVT PIB – Livestock Sector to have Digital Footprints: Bharat Pashudhan / NDIM [pib.gov.in/PressReleases.nsf/\(Symbol\)/PIB-2025-100](https://pib.gov.in/PressReleases.nsf/(Symbol)/PIB-2025-100)
- GOVT PIB – National Digital Livestock Mission: Government of India / PIB [pib.gov.in/PressReleases.nsf/\(Symbol\)/PIB-2025-89](https://pib.gov.in/PressReleases.nsf/(Symbol)/PIB-2025-89)
- RESEARCH ICAR-NIVEDI – NADRES v2, AI-enabled livestock disease forewarning ICAR-NIVEDI https://res.in/nadres_v2/
- RESEARCH NADRES – Livestock Disease Risk Forewarning Methodology bulletins- NADRES / ICAR-NIVEDI https://res.in/nadres_v2/livestockdiseasesforecast
- POLICY PRS India – Lumpy Skin Disease in Cattle : Report Summary india.prs.gov.in/report-summaries/lumpy-skin-disease-in-cattle
- GOVT DAHD – State-wise LSD deaths (Lok Sabha reply LS872) : Department of Animal Husbandry & Dairying dahd.gov.in/sites/default/files/2024-10/LS872.pdf
- SERVICE EMRI – 1962 Animal Mobile Medical Ambulance : Emergency Management and Research Institute emri.in/3592-animal-mobile-medical-ambulance-2/

Comparison with Existing Solutions:

Features	Pashu Aarogya (Our Solution)	Bharat Pashudhan	NADE-RS	Pashudhan Sanjvani
Animal & Vaccination records	✓	✓	✗	○
Disease forecasting / risk maps	✓	✗	✓	✗
Farmer - first symptom reporting	✓	○	○	✓
Ai-assisted case risk triage	✓	○	○	○
Local language guided reporting	✓	○	✗	○
Cluster Detection from farmer reports	✓	○	○	✗
Nearby veterinary/ service mapping	✓	○	○	✓
Farmer → Vet → Officer workflow	✓	✓	○	✓
Unified alerts + response dashboard	✓	✓	○	○
Prevention + response in one workflow	✓	✓	✓	✓

○ → LIMITED

6 · Research and References

- NADE-RS → NADRES · "HCKATHON" typo
- All ten rows green – concede two to Bharat Pashudhan and 1962
- Add the 2009 Act as a reference (the 83-disease baseline)
- Add the source for the ₹20,000 Cr figure

What the judges will ask, and what we say

QUESTION	WHAT THE DECK SETS US UP TO SAY TODAY	WHAT WE SHOULD SAY INSTEAD
"Show me the XGBoost model."	Nothing — it does not exist. Every other claim is now in doubt.	"We deliberately did not use a black box. Triage is transparent weighted scoring over rules the department can edit, and the cluster detection is a Kulldorff scan statistic with a real p-value. Here is the p-value on screen."
"How is this different from 1962 or Bharat Pashudhan?"	A feature table where we are better at everything.	"They treat one animal at a time and hold the lifetime record. Nobody aggregates that into village-level early warning. We are the layer on top, and we read their data by Pashu Aadhaar."
"How do you know it reduces reporting time?"	No measurement is shown anywhere in the deck.	"We ask the farmer when the animal fell ill, so the delay is measured, not assumed. Median has moved 6.4 → 1.5 days as adoption rises. It is on the dashboard, live."
"Is this data real?"	Slides imply a completed 20-farmer pilot.	"The history is simulated and shaped like the 2022 LSD wave — we will say so every time. The farmers registering on this screen right now are real."
"Only 17 diseases?"	"17+" with no baseline.	"13 of the 83 notifiable under the 2009 Act, plus 4 production diseases farmers actually call about. Name one you want and I will add it now — it is 20 lines of YAML and a reload."
"What if you are wrong about the disease?"	Not addressed on any slide.	"We never diagnose. We output a suspected category, a severity band and a route; a registered vet confirms. That is enforced in the architecture."
"Who pays for it, and who runs it?"	Cost model is there, but no numbers.	"Five of six features need no GPU. One app server plus Postgres serves a pilot district on any MeitY-empanelled cloud, and the department owns the data and the rules."

One rule for the whole team. Only claim what one of us has personally checked. If it came from a website, say "their page describes...". Never present simulated data as real — volunteer that it is simulated before anyone asks. Teams lose finals on one over-claim, not on a missing feature.

Work order — credibility first, polish last

PRIO	TASK	SLIDE	EFFORT	DONE WHEN...	OWNER
P0	Replace the false stack and ML claims with the real ones (copy on page 2)	2, 3, 4	1 hour	No slide names a library that is not in the repo	_____
P0	Fix INAPH, NADRES, HCKATHON, Traige and the cut-off text box	2, 3, 4, 5, 6	10 min	Spell-check passes; nothing overflows	_____
P0	Settle the pilot number — 18 or 20, and completed or scheduled	2, 4	10 min	One number, one tense, everywhere	_____
P0	Add the reporting-delay metric 6.4 → 1.5 days	5	20 min	An expected outcome from the PS carries a number	_____
P1	Reframe as the layer on top of 1962 / Bharat Pashudhan; concede two rows	2, 6	30 min	The table shows two rows where they win	_____
P1	Add the safety rule — suspected category, never a diagnosis	2 or 5	10 min	The liability answer is on a slide	_____
P1	Reframe the disease count against the 83 notifiable	2, 6	20 min	"13 of 83 + 4" replaces "17+"	_____
P1	Register 20 real farmers so the pilot claim is simply true	—	1 day	The live wall reads 20 / 20	_____
P1	Cite the ₹20,000 Cr figure; add the 2009 Act as a reference	5, 6	20 min	Every number on the deck has a source	_____
P2	Thin slide 2 — Uniqueness down to three points; raise body text size	2	30 min	Readable from the back of the room	_____
P2	Add a QR code to the live app on the title slide	1	15 min	A judge can open it on their own phone	_____
P2	Measure breed-model accuracy, or soften the 95% claim	2	Half day	A confusion matrix exists, or the number is gone	_____

If you only do one thing: delete every library name we do not run, and replace the DBSCAN line with the Kulldorff scan statistic. It takes an hour, it makes the deck more impressive rather than less, and it removes the single question that could unravel the whole presentation.